

MediAI Expert

AI-Powered Medical Diagnosis Expert System

1. Project Overview

MediAI Expert is a web-based artificial intelligence application designed to simulate the decision-making ability of a human medical expert. The system utilizes foundational symbolic AI concepts—specifically **Forward Chaining** and **Backward Chaining**—to diagnose common ailments based on patient symptoms.

It features a modern, interactive, and clinical user interface, ensuring a seamless experience for users while maintaining robust backend logic for rule-based inference.

2. System Architecture

The application follows a classic Client-Server architecture:

- **Frontend (Client-Side):** Built with HTML5, CSS3, and Vanilla JavaScript. It provides a Single Page Application (SPA) experience using asynchronous `fetch()` API calls to prevent page reloads during the diagnostic process. The UI leverages a "glassmorphism" aesthetic with an interactive canvas background to create a premium, modern feel.
- **Backend (Server-Side):** Developed using Python and the Flask framework. The backend maintains the Expert System's Knowledge Base, exposes RESTful API endpoints (`/api/forward` and `/api/backward`), and handles the inference engine logic.
- **Persistence Layer:** Diagnostic sessions are logged locally into text files within an `output/` directory, uniquely named after the registered patient, establishing an audit trail of the diagnostic reasoning.

3. Core AI Concepts Implemented

The project acts as an educational and practical implementation of an Expert System, leveraging two primary inference mechanisms:

3.1 Knowledge Base (KB)

The system's Knowledge Base consists of a predefined set of rules mapping symptoms to specific diseases:

- **Flu:** Fever, Cough
- **Dengue:** Fever, Headache
- **Food Poisoning:** Stomach Pain, Vomiting
- **Common Cold:** Cold, Sneezing

3.2 Forward Chaining (Data-Driven Inference)

In this mode, the system starts with the available data (symptoms) and applies inference rules to extract more data until a goal (diagnosis) is reached.

- **Workflow:** The user selects all symptoms they are currently experiencing. The backend evaluates these symptoms against the conditions of all known rules.
- **Logic:** If the selected symptoms form a superset or exact match of a rule's conditions (e.g., `{"fever", "cough"}.issubset(user_symptoms)`), the system confidently concludes the patient has the corresponding disease.

3.3 Backward Chaining (Goal-Driven Inference)

In this mode, the system starts with a hypothesis (a suspected disease) and works backward to see if the available evidence (symptoms) supports that hypothesis.

- **Workflow:** The clinical user inputs a suspected disease and verifies the presence of specific symptoms.
- **Logic:** The backend retrieves the required symptoms for the hypothesized disease from the Knowledge Base and checks if the user's provided symptoms satisfy those requirements.

4. Key Features & User Flow

1. **Patient Registration:** Every session begins with the user entering the patient's name. This personalizes the interaction and links the diagnostic session log to a specific individual.
2. **Mode Selection:** Users are prompted to select their preferred diagnostic methodology (Forward or Backward chaining) based on the context of the consultation.
3. **Interactive Symptom Grid:** A clean, accessible grid allows users to toggle symptoms on and off.
4. **Real-time Processing Overlay:** A subtle loading state appears during API calls to provide immediate user feedback while the inference engine processes the rules.
5. **Comprehensive Diagnostic Report:** The final result view displays not just the final diagnosis, but a fully generated logical step-by-step report of the inference process.
6. **Session Logging:** A timestamped log of the entire diagnostic process (including the chosen mode, symptoms, hypotheses, and conclusions) is written to `output/[Patient Name].txt` for record-keeping.

5. UI/UX Design Aesthetics

The frontend was designed with a focus on modern web standards and high-end aesthetics:

- **Glassmorphism:** The main application view operates within a frosted "glass card" that floats over the background.
- **Dynamic Background:** An HTML5 `<canvas>` provides a subtle, interactive background animation, lending a high-tech "AI" feel to the application.
- **Typography & Color Scheme:** Clean sans-serif fonts (Inter) paired with a clinical color palette consisting of varied blues and pure whites to establish trust and professionalism.

6. Conclusion

The **MediAI Expert** effectively models a simple but structurally sound AI Expert System. It successfully abstracts complex logical routing behind a highly polished, user-friendly interface, making it an excellent demonstration of applied logical inference and full-stack web development.